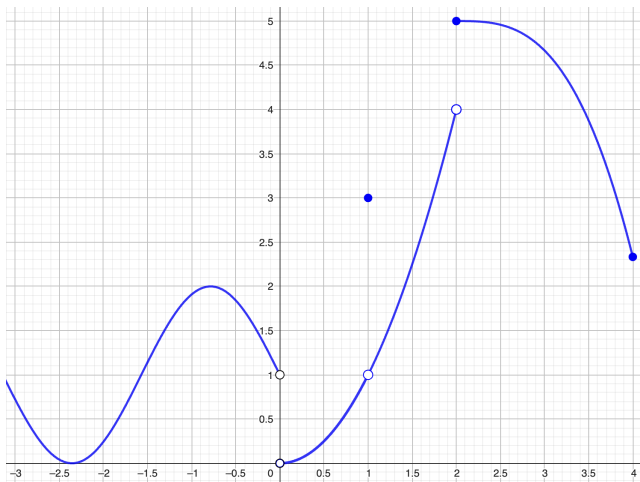


Tutorial - Week 3

TG 11 - TG 12

1. For the function f whose graph is given, state the value of each quantity if it exists. If it does not exist, explain why.



(a) $\lim_{x \rightarrow 0^-} f(x) =$

(b) $\lim_{x \rightarrow 0^+} f(x) =$

(c) $f(1) =$

(d) $\lim_{x \rightarrow 1^-} f(x) =$

(e) $\lim_{x \rightarrow 1^+} f(x) =$

(f) $\lim_{x \rightarrow 1} f(x) =$

(g) $f(2) =$

(h) $\lim_{x \rightarrow 2^-} f(x) =$

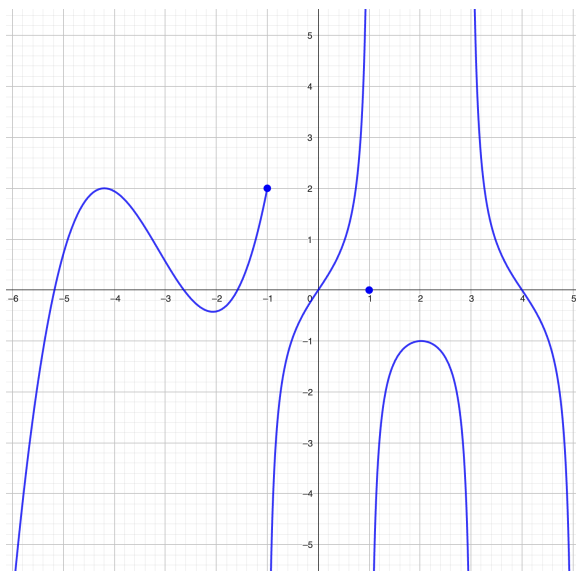
(i) $\lim_{x \rightarrow 2^+} f(x) =$

(j) $\lim_{x \rightarrow 2} f(x) =$

(k) $f(4) \approx$

(l) $\lim_{x \rightarrow 4^-} f(x) \approx$

2. For the function f whose graph is shown, calculate the following:



(a) $\lim_{x \rightarrow -6^+} f(x) =$

(c) $\lim_{x \rightarrow -1^-} f(x) =$

(e) $\lim_{x \rightarrow -1} f(x) =$

(b) $f(-1) =$

(d) $\lim_{x \rightarrow -1^+} f(x) =$

(f) $f(1) =$

(g) $\lim_{x \rightarrow 1^-} f(x) =$

(i) $\lim_{x \rightarrow 1} f(x) =$

(k) $\lim_{x \rightarrow 3^+} f(x) =$

(h) $\lim_{x \rightarrow 1^+} f(x) =$

(j) $\lim_{x \rightarrow 3^-} f(x) =$

(l) the equations of the vertical asymptotes.

3. Sketch the graph of a function f that satisfies all the given conditions.

(a) $\lim_{x \rightarrow -4^+} f(x) = -\infty$.

(i) $f(0) = -2$.

(b) $\lim_{x \rightarrow -2^-} f(x) = +\infty$.

(j) $\lim_{x \rightarrow 0^-} f(x) = -\infty$.

(c) $\lim_{x \rightarrow -2^+} f(x) = -\infty$.

(k) $\lim_{x \rightarrow 0^+} f(x) = -2$.

(d) $f(-2) = 1$.

(l) f is constant in $(0, 1)$.

(e) f is increasing in $(-4, -2) \cup (-2, -1)$.

(m) $f(1) = 0$.

(f) -1 is not in the domain of f .

(n) f is constant in $(1, 4]$.

(g) $\lim_{x \rightarrow -1} f(x) = 2$.

(o) $\lim_{x \rightarrow 2^+} f(x) = 2$.

(h) f is decreasing in $(-1, 0)$.

4. Calculate the following limits. For those that do not exist, determine if there exist one-sided limits.

(a) $\lim_{x \rightarrow -6} \frac{-20\sqrt{x+8}}{|2x+2|}$.

(d) $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{|x|} \right)$.

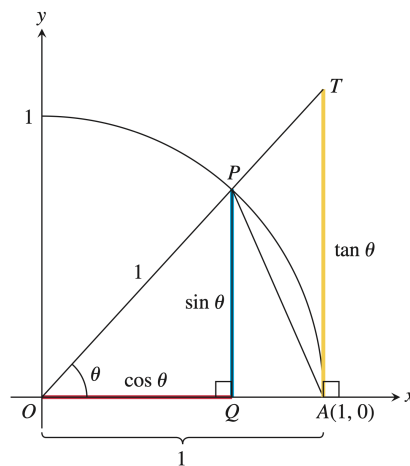
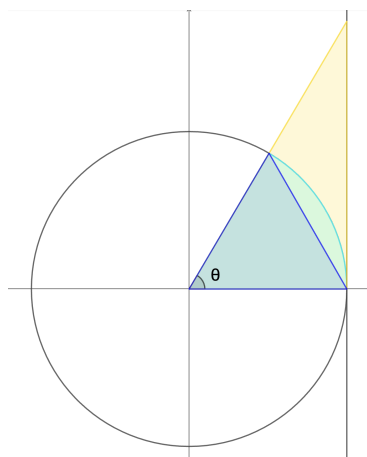
(b) $\lim_{x \rightarrow -2} \frac{x^3 + 3x^2 + 2x}{x^2 - 4}$.

(e) $\lim_{x \rightarrow 2} \frac{x-2}{\sqrt{2x-3}-1}$.

(c) $\lim_{x \rightarrow 0} \sin \left(\frac{\pi x}{|x|} \right)$.

(f) $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x}$.

5. Using that $\sin x \leq x \leq \tan x \ \forall x \in (0, \frac{\pi}{2})$, show that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$.



6. Given $\varepsilon = \frac{1}{5}$, find δ such that:

(a) $|x-1| < \delta$ implies $|\sqrt{x}-1| < \varepsilon$.

(b) $|x-3| < \delta$ implies $|2x^2-4x-6| < \varepsilon$.

(c) $|x-2| < \delta$ implies $|\frac{1}{x^2} - \frac{1}{4}| < \varepsilon$.

7. Using the definition of limits prove:

- (a) $\lim_{x \rightarrow a} c = c$ for any $a \in \mathbb{R}$.
- (b) $\lim_{x \rightarrow a} x = a$ for any $a \in \mathbb{R}$.
- (c) $\lim_{x \rightarrow a} x^2 = a^2$ for any $a \in \mathbb{R}$.
- (d) $\lim_{x \rightarrow 0} \left(x^2 \sin \left(\frac{a}{x^n} \right) \right) = 0$ for any $a \in \mathbb{R}$ and $n \in \mathbb{N}$.
- (e) $\lim_{x \rightarrow 0} \left(x^n \tan^{-1}(\sin ax) \right) = 0$ for any $a \in \mathbb{R}$ and $n \in \mathbb{N}$.
- (f) $\lim_{x \rightarrow 0^+} \frac{1}{x} = \infty$.
- (g) $\lim_{x \rightarrow 0^+} \ln x = -\infty$.

8. Let a and c be two real numbers, and let f and g be functions such that $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = L'$. Prove the following statements.

- (a) $\lim_{x \rightarrow a} cf(x) = cL$.
- (b) $\lim_{x \rightarrow a} f(x)g(x) = LL'$.

9. Use Exercises 7. and 8. to show that $\lim_{x \rightarrow a} p(x) = p(a)$ for any polynomial p in $\mathbb{R}[x]$ and a in $\mathbb{R}$.

10. Suppose that $\lim_{x \rightarrow a} f(x) = \infty$ and $\lim_{x \rightarrow a} g(x) = c$, where c is a real number. Prove each statement.

- (a) $\lim_{x \rightarrow a} (f(x) + g(x)) = \infty$.
- (b) $\lim_{x \rightarrow a} f(x)g(x) = \infty$ if $c > 0$.
- (c) $\lim_{x \rightarrow a} f(x)g(x) = -\infty$ if $c < 0$.